

Identifying Service Boundaries

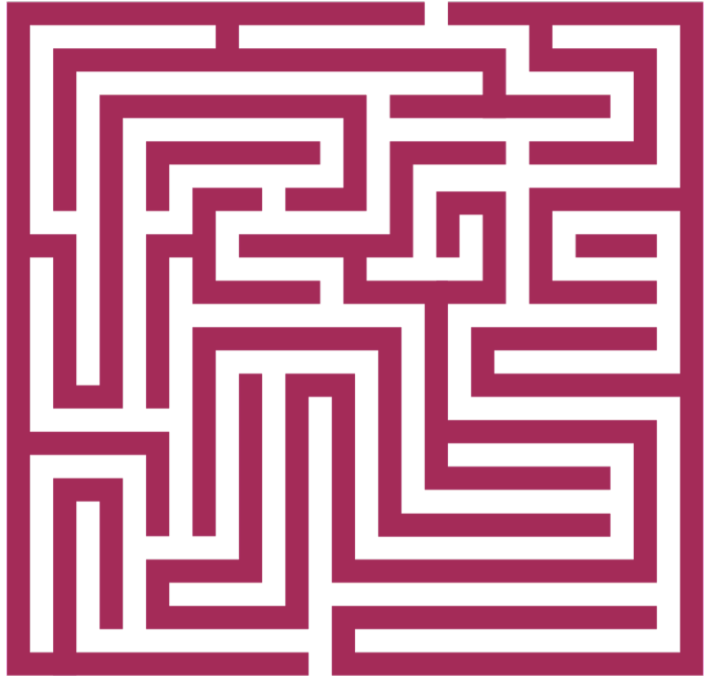


Michael L Perry

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Microservices

Service-oriented architecture

Complexity

Advantage?



Why Separate Services?

Business reasons

Different incentives
Different decision makers

Technical reasons

Different data stores
Different release schedules
Different usage patterns





Uses many tools

Identify service boundaries





Organizes service design workshop

Invites stakeholders

Event storming





Alberto Brandolini

eventstorming.com



Scoping and Organizing .NET Microservices Using Event Storming

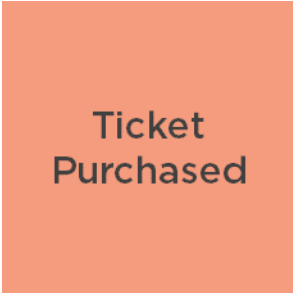


Rag Dhiman

@ragdhiman



Events



Ticket
Purchased

Domain events

Past tense verbs

Orange stickies

Timeline



Event Timeline

Venue
Booked

Ticket
Purchased



Event Timeline

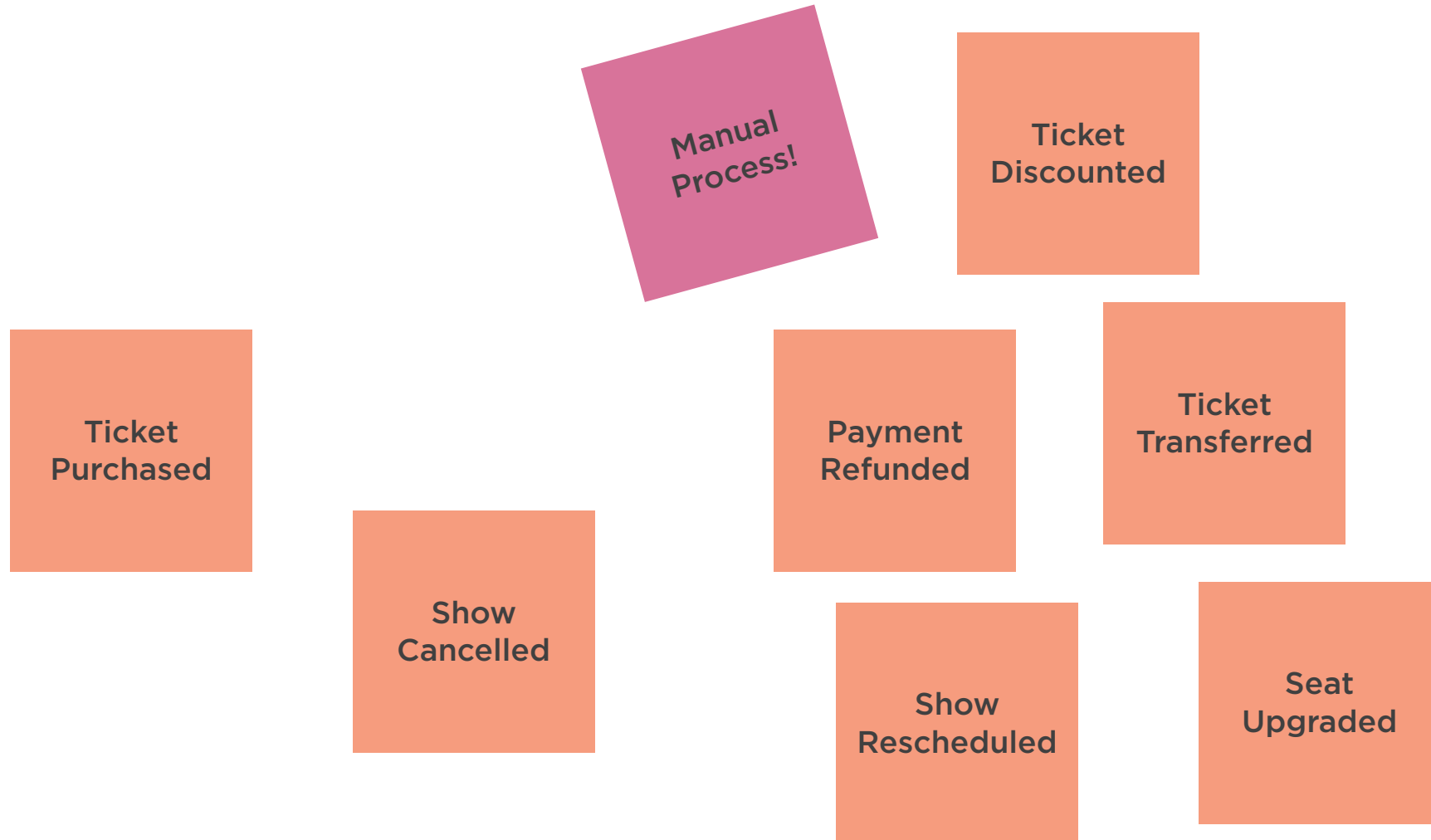
**Contract
Signed**

**Venue
Booked**

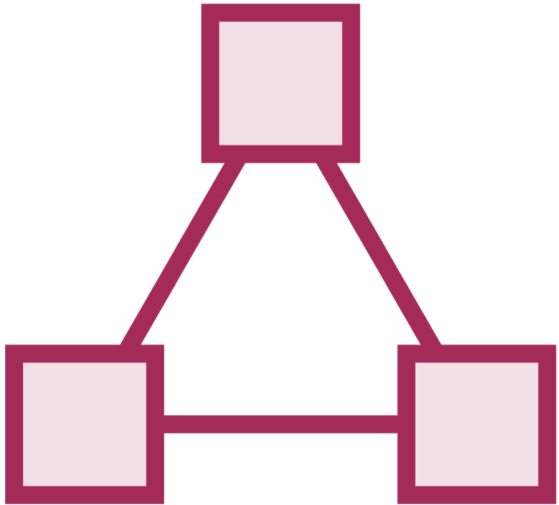
**Ticket
Purchased**



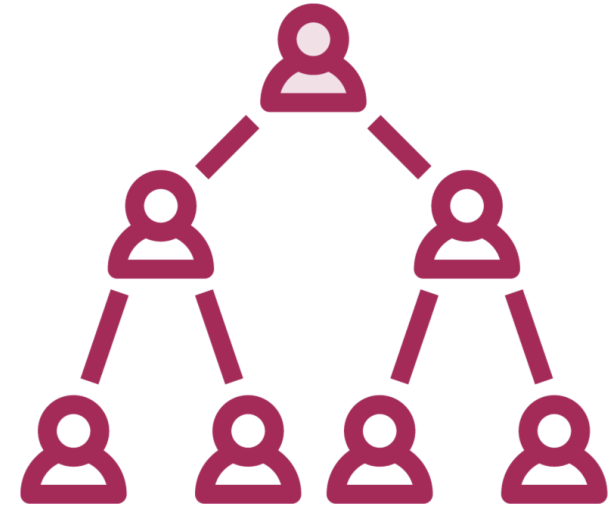
Pain Points



Pain Points



Technical problems



Organizational problems

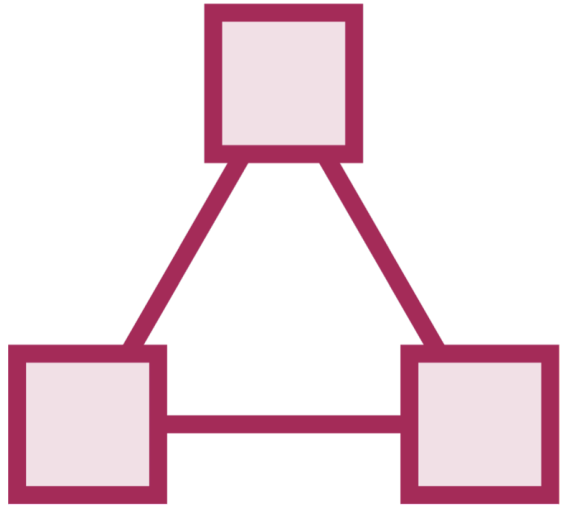


“Any organization that designs a system will produce a design whose structure is a copy of the organization’s communication structure.”

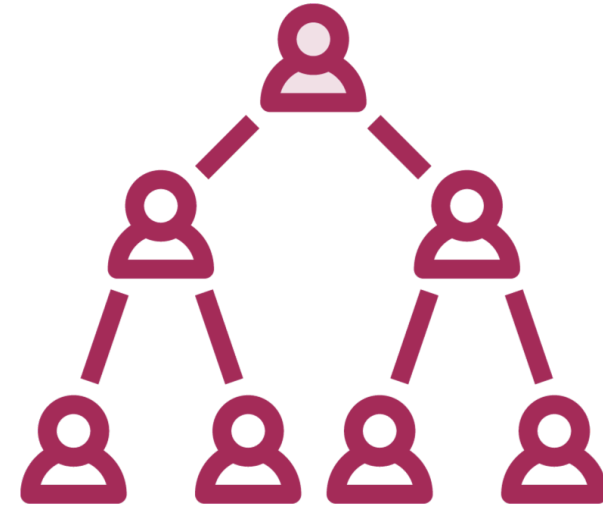
Melvin E. Conway



Communication Structure



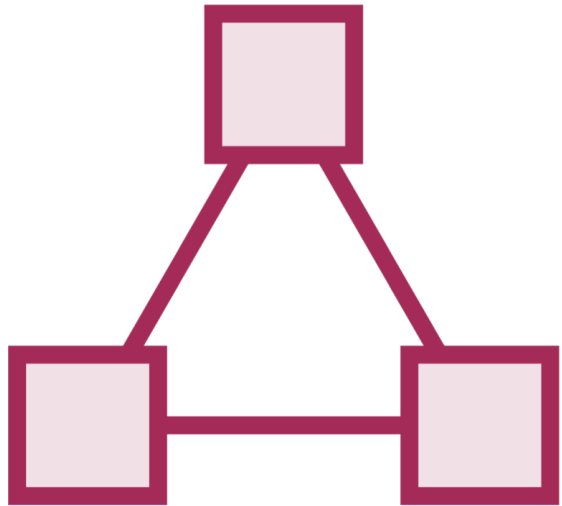
Technical structure



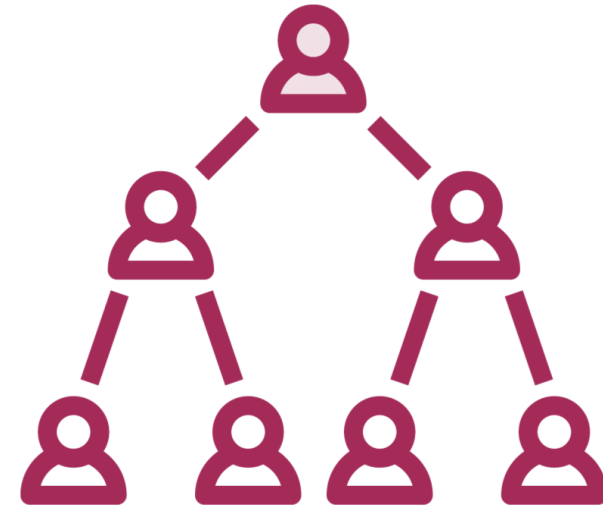
Organizational structure



Inverse Conway Maneuver



Technical structure

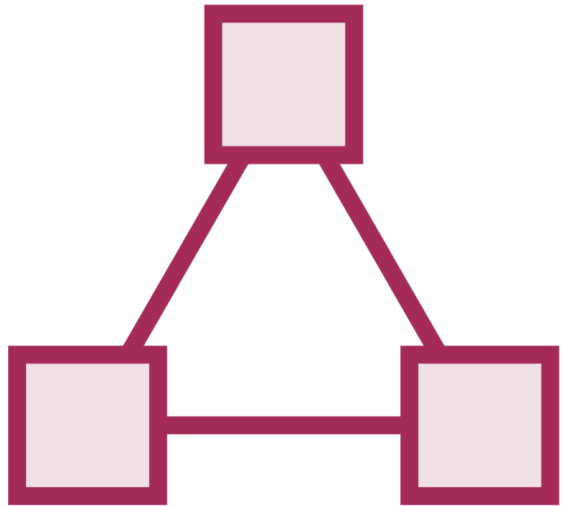


Organizational structure

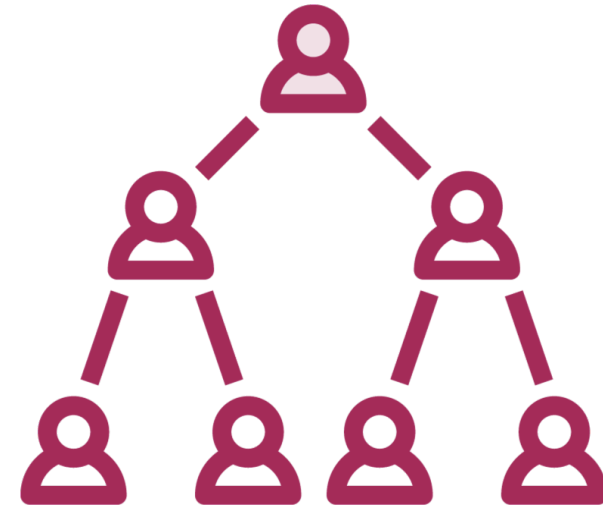
Is this backwards?
Software serves
organization
Should organization
change?



Structure is Coupled

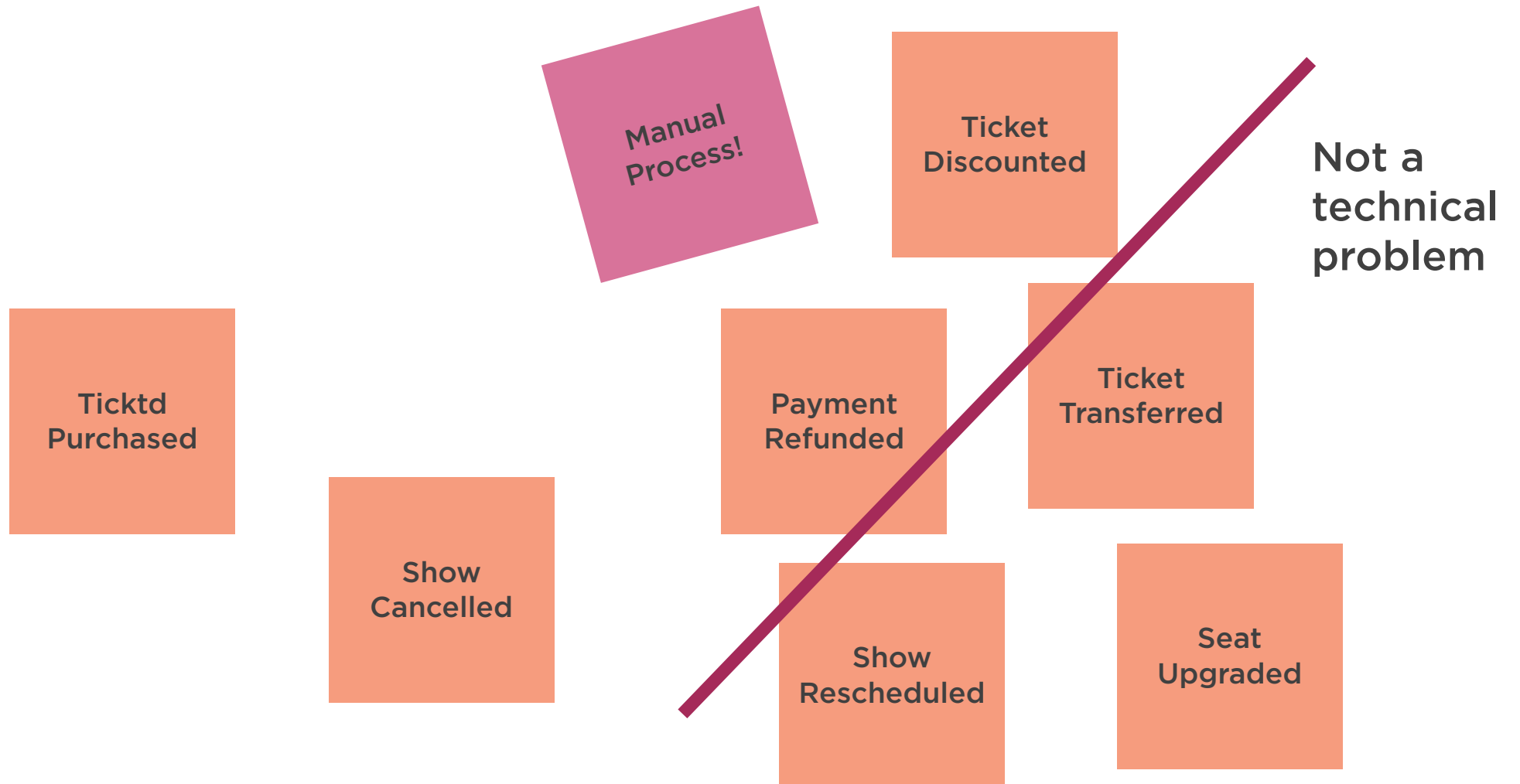


Technical structure



Organizational structure

Pain Points



Incentives

Sales

Increase volumes

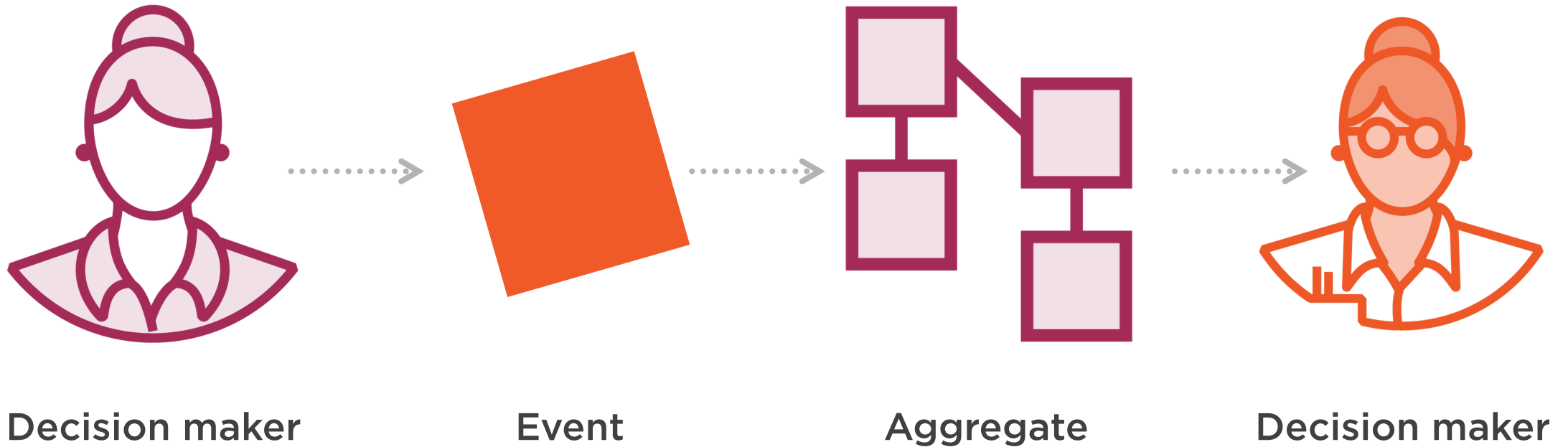
Remediation

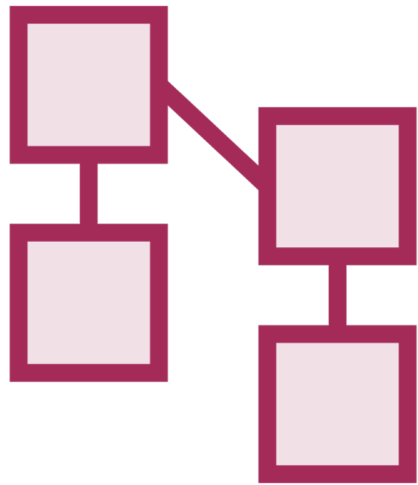
Reduce cost

- by reducing volume
- or by postponing sales



Decision Propagation

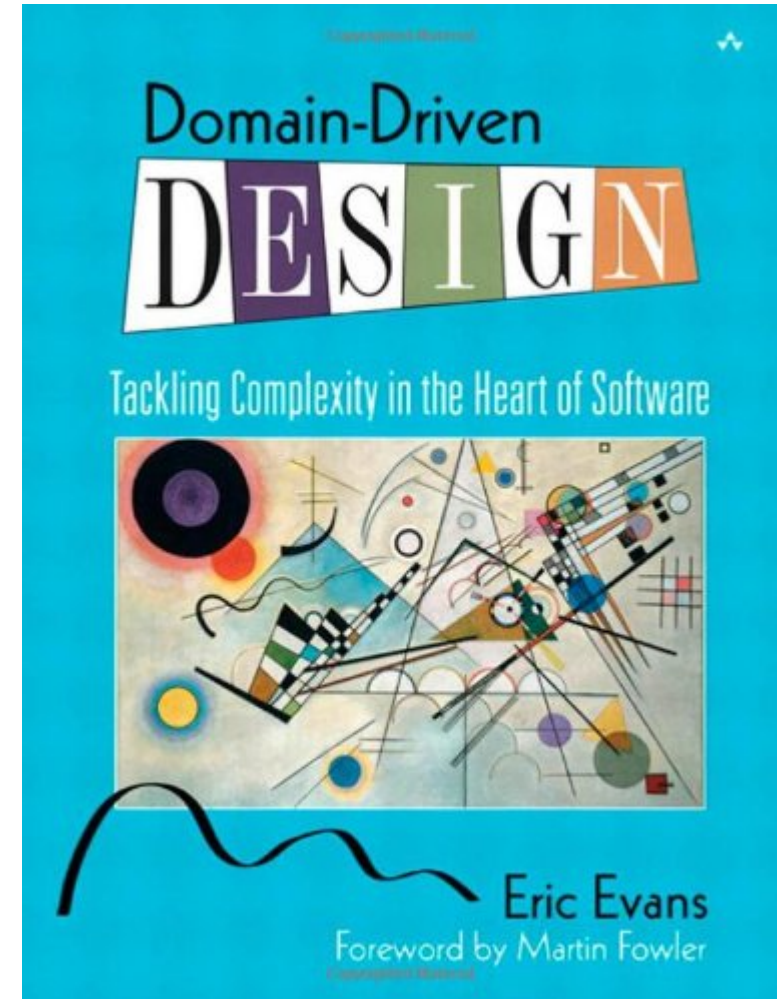




Aggregate



Eric Evans



Domain Driven Design in Practice



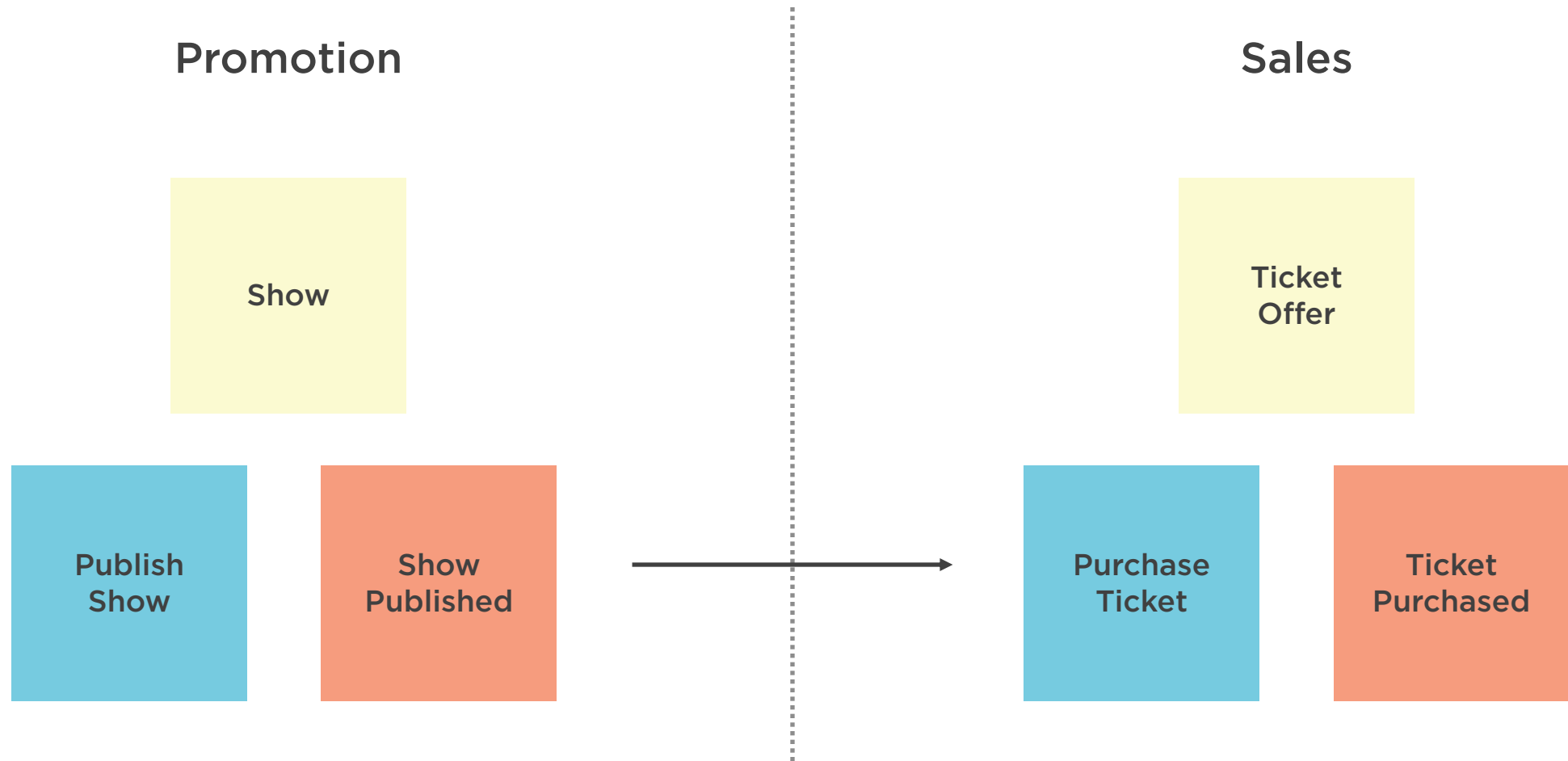
Vladimir Khorikov

@vkhorikov

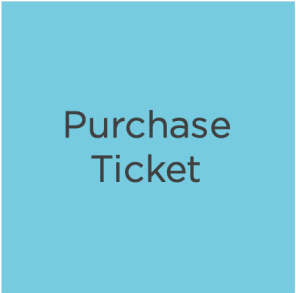
www.enterprisecraftsmanship.com



Context Boundary



Commands



Purchase
Ticket

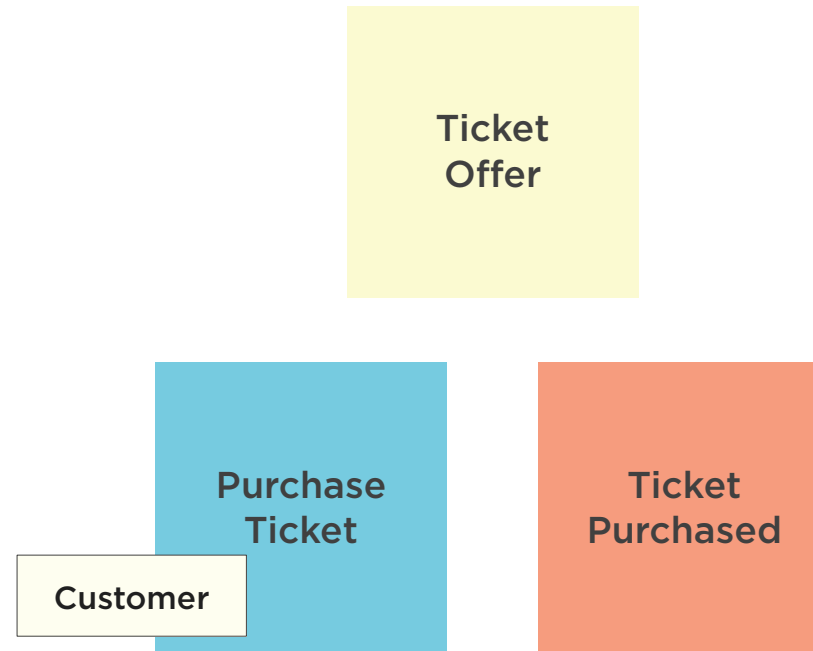
Imperative verbs

Blue stickies

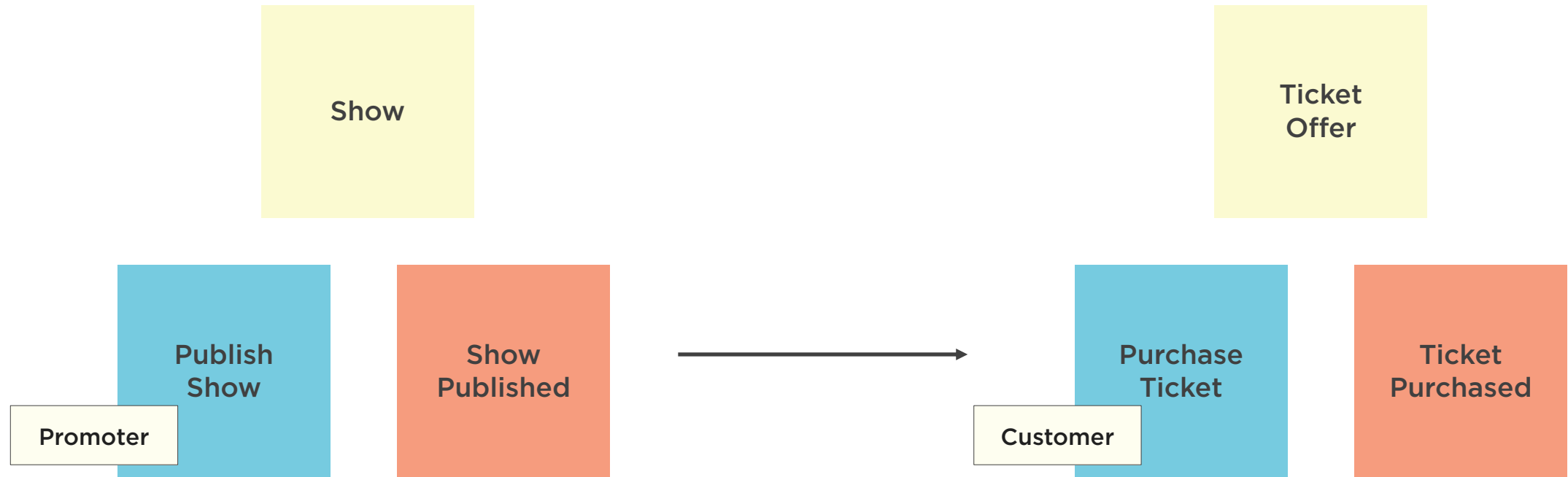
Paired with events



Who and What

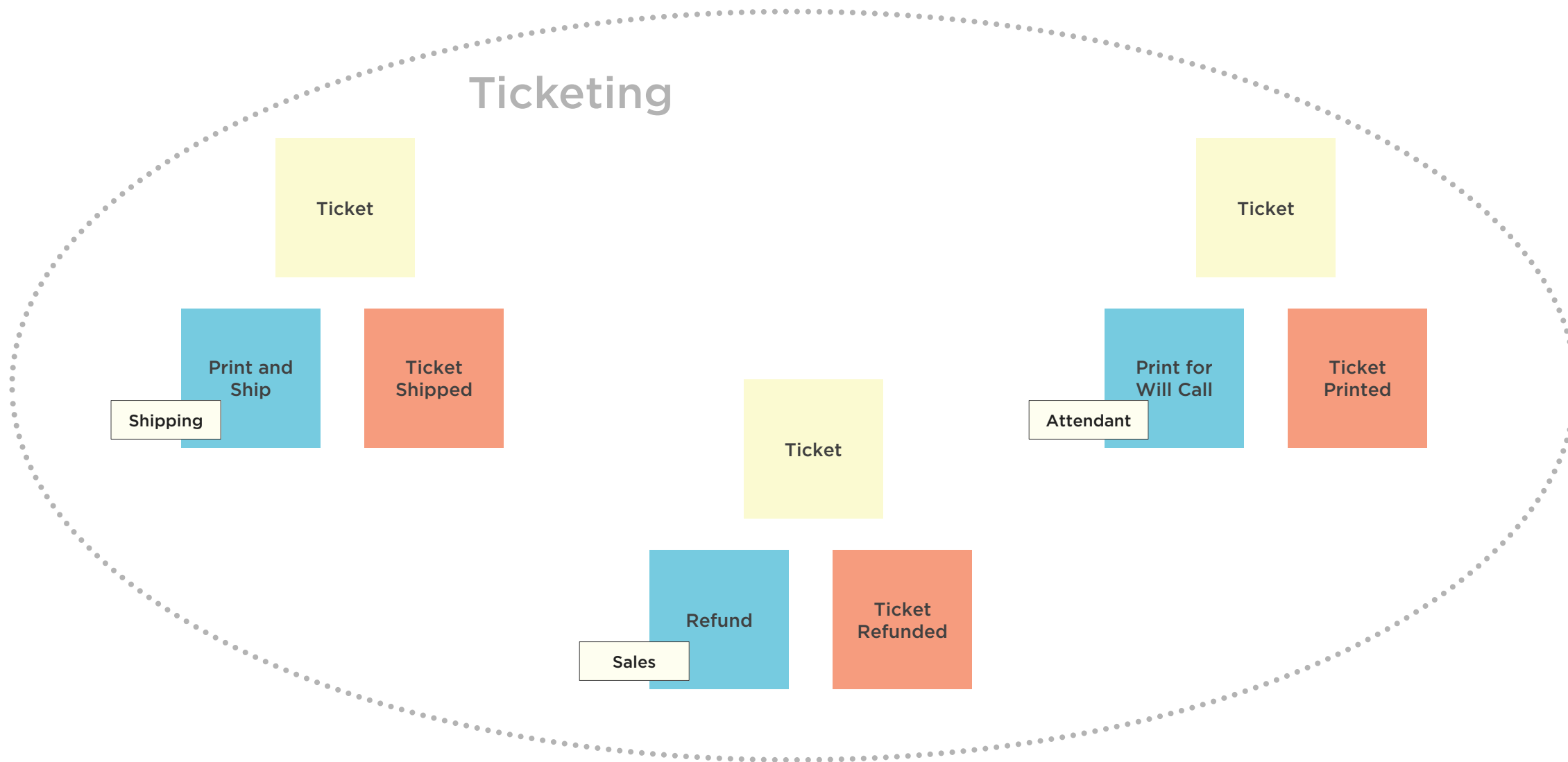


Who and What



Reused Aggregates

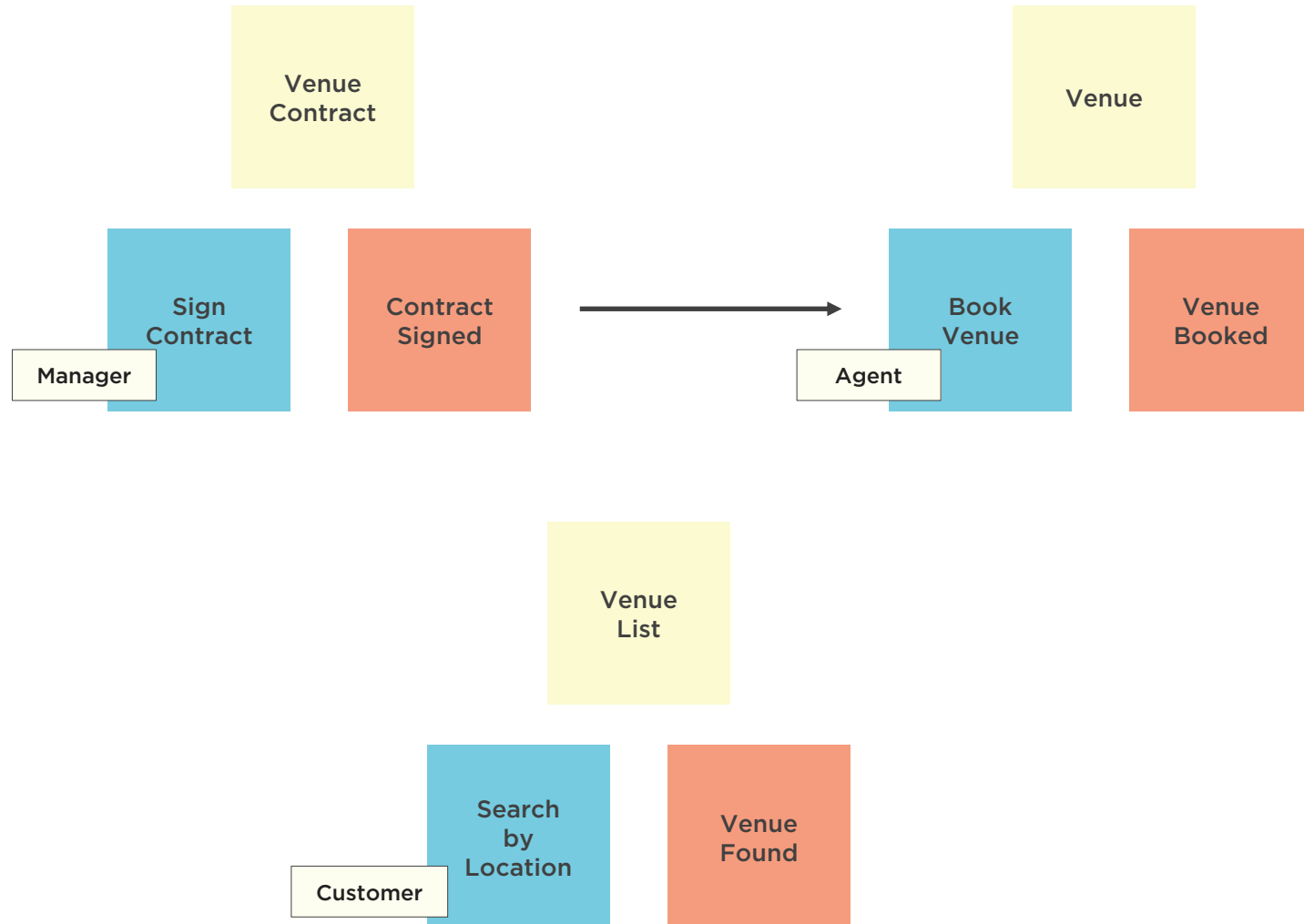
Ticketing



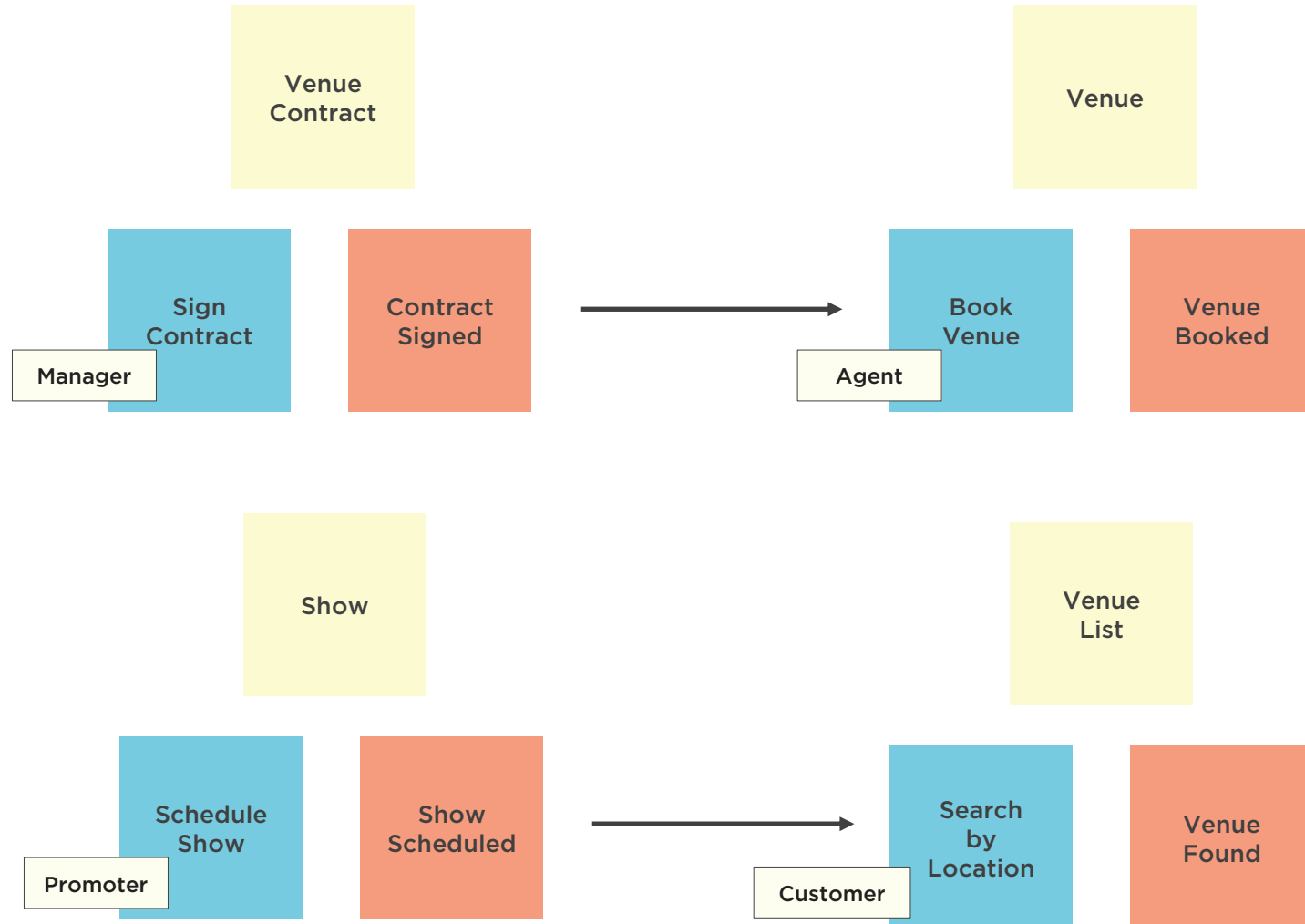
Similarly Named Aggregates



Similarly Named Aggregates



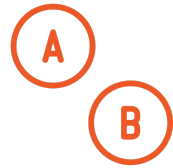
Similarly Named Aggregates



Distinguishing Aggregates



Definition



Differentiation



Identity



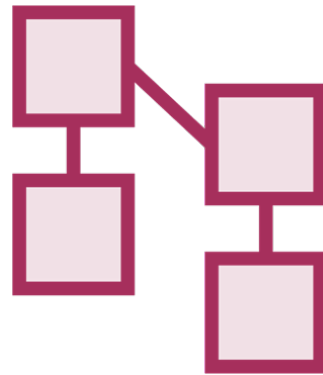
Events



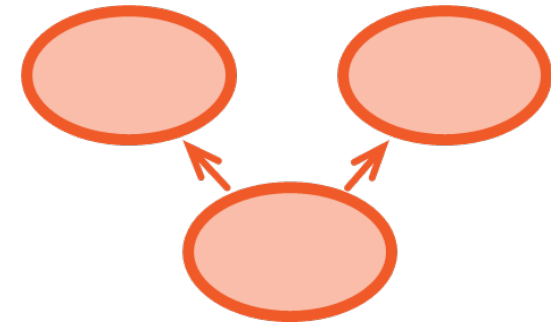
Service Design Workshop



Event



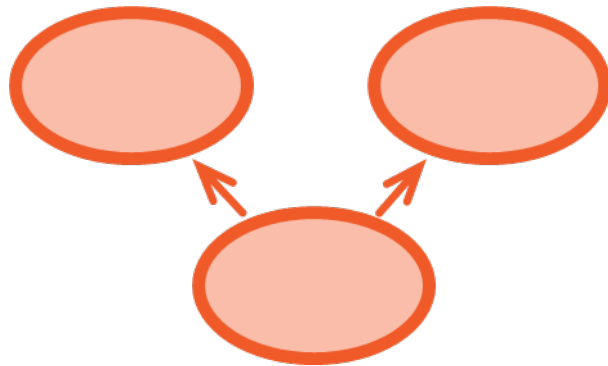
Aggregates



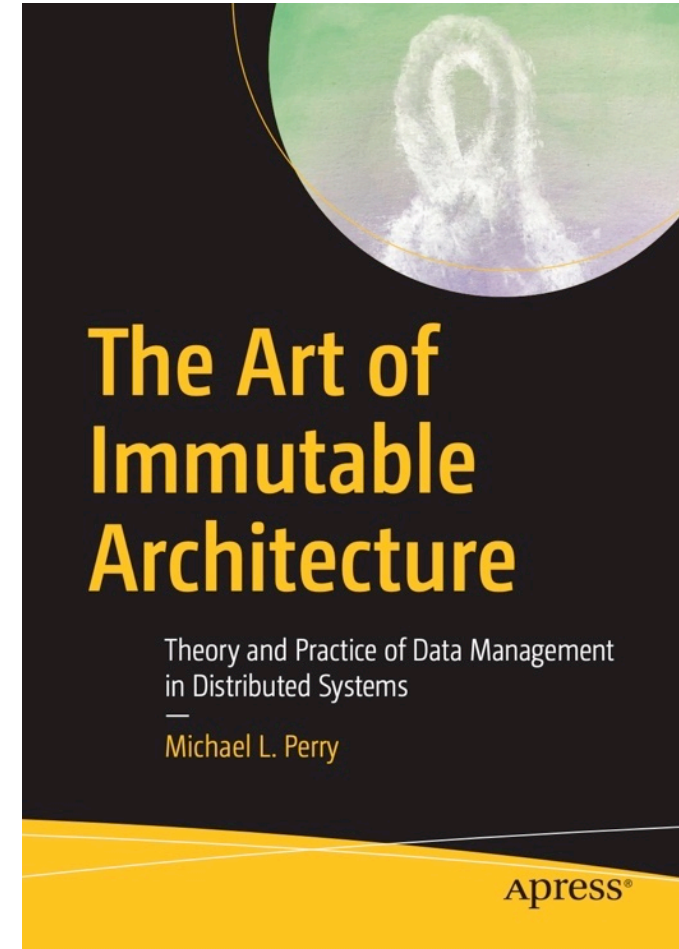
Decisions



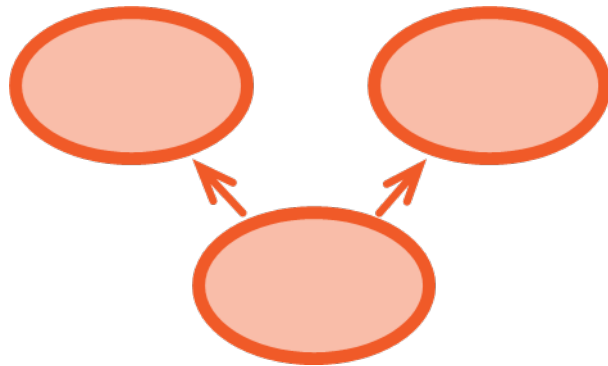
Historical Modeling



Michael L Perry



Historical Modeling



Historical fact

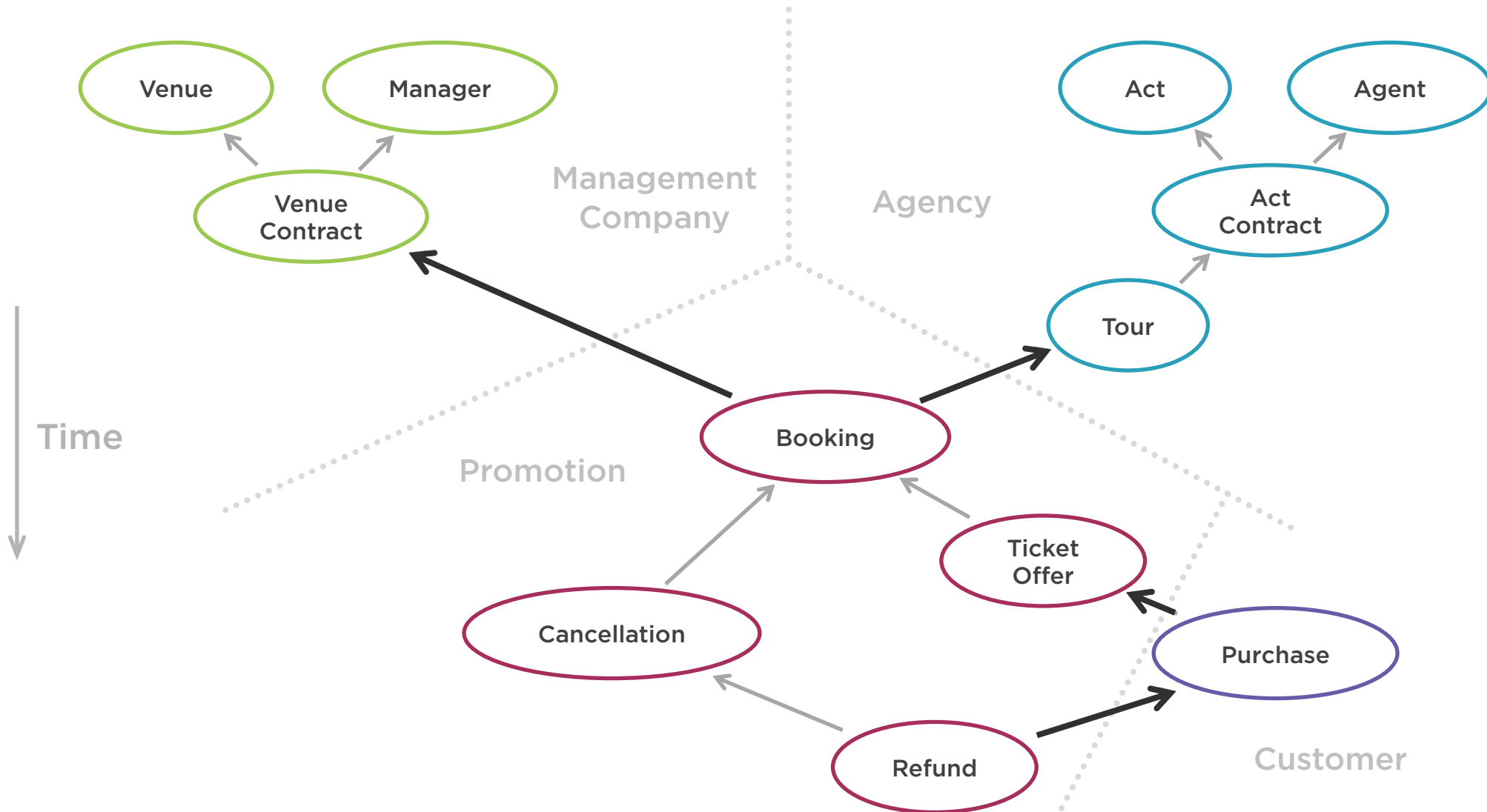
Refers to predecessors

Decisions that have occurred

Which decisions led to new decisions



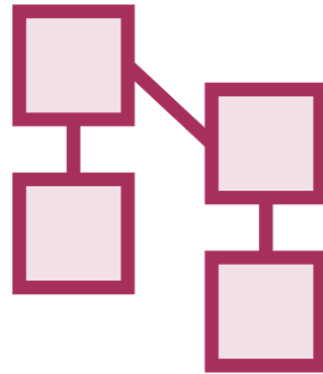
Historical Model



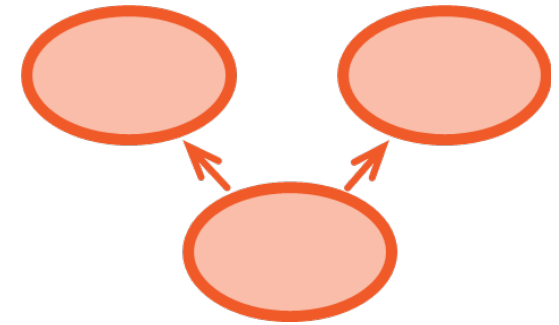
Service Design Workshop



Event
Storming



Domain
Model



Historical
Model



Why Separate Services?

Business reasons

Different incentives

Different decision makers

Technical reasons

Different data stores

Different release schedules

Different usage patterns



Service Design

Core Services

- Sales
- Billing
- Customer Service
- Remediation

Integration Services

- Agency Relations
- Promotion
- Venue Management

Technical Separation

- Search

Business Separation

- Asynchronous Communications
- Synchronous Communications

Proxy

Interactions:

- Tour (Agency Relations to Promotion)
- Ticket Offer (Sales to Search)
- Ticket Purchased (Sales to Billing)
- Show Scheduled (Promotion to Search)
- Show Cancelled (Promotion to Search)
- Venue Contract (Venue Management to Promotion)
- Charge (Billing to Remediation)
- Charge Back (Remediation to Billing)
- Remedy Offer (Remediation to Customer Service)



Service design informs

- Software
- Projects
- Teams

Attach each service to a business unit

Recommend organizational changes

Continue to refine service design



Up Next:

Invoking Business Processes

